

IBM HR Analytics Dashboard

1. Project Overview

This project was developed as part of the **Advanced Level Data Analyst Internship at ShadowFox**. The objective was to transform an HR dataset into an interactive, executive-style dashboard using **Microsoft Power BI**.

The **IBM HR Analytics dataset** was used to analyze employee demographics, job roles, job satisfaction, tenure, overtime, departmental distribution, and employee attrition.

The final dashboard provides a single-page view of important workforce metrics and allows users to interactively explore employee and attrition patterns. The Advanced task emphasizes KPI selection, dashboard planning, interactive reporting, filters/slicers, business-focused storytelling, and decision-oriented insights.

2. Dataset

The dataset contains employee-level HR information covering areas such as:

- Employee demographics
- Department
- Job Role
- Job Level
- Age
- Gender
- Monthly Income
- Years at Company
- Job Satisfaction
- Overtime
- Attrition

The dataset provides sufficient information to study workforce composition and employee attrition patterns.

3. Data Preparation

Before developing the dashboard, the dataset was inspected and prepared for analysis.

The data preparation process included:

- Understanding the dataset structure

- Checking data types
- Checking missing values
- Checking duplicate records
- Reviewing categorical values
- Checking numerical fields
- Preparing fields for visualization
- Creating useful groups/categories for analysis where required
- Preparing the final dataset for Power BI

The prepared dataset was then imported into Power BI for dashboard development.

4. Dashboard KPIs

The dashboard contains six main KPI cards:

Total Employees

Provides an overview of the employee population represented in the dataset.

Attrition Count

Shows the employee attrition measure used in the dashboard.

Average Job Satisfaction

Displays the average job satisfaction level of employees.

Average Age

Shows the average age of employees.

Average Monthly Income

Provides an overview of average employee monthly income.

Average Years at Company

Shows the average employee tenure within the organization.

These KPIs provide an executive-level summary before users move into detailed visual analysis.

5. Dashboard Visualizations

The dashboard contains the following visual analyses.

5.1 Attrition by Department

A bar chart is used to compare employee attrition across departments.

This helps identify differences in attrition between organizational units.

5.2 Job Role by Attrition

This visualization compares job roles according to attrition.

It helps identify how employee attrition is distributed across different roles.

5.3 OverTime by Attrition

A donut chart compares overtime-related employee groups with attrition.

This provides a quick visual understanding of the relationship between overtime and employee attrition.

5.4 Age Group by Attrition

Employees are grouped according to age categories and compared using attrition.

This helps identify differences in attrition across employee age groups.

5.5 Average Years at Company by Attrition

This chart compares average company tenure between employees according to attrition status.

It provides insight into the relationship between employee tenure and attrition.

5.6 Employee Number Distribution by Department

A department-level distribution of employee numbers is included to provide an additional view of the workforce across departments.

5.7 Average Job Satisfaction by Department

This bar chart compares average job satisfaction across departments.

It helps identify differences in employee satisfaction between organizational units.

6. Interactive Slicers

The dashboard includes interactive filters that allow users to explore the data dynamically.

The available slicers include:

- **Department**
- **Age Group**
- **Job Role**
- **Job Level**
- **OverTime**

These slicers allow users to focus the dashboard on particular employee groups and examine how the selected characteristics affect the displayed analysis.

For example, a user can select a specific job role or department and examine the corresponding attrition, tenure, overtime, and satisfaction information.

The use of slicers satisfies the interactive reporting requirement of the Advanced task.

7. Key Insights

The dashboard enables several important HR observations.

Departmental Attrition

Attrition varies across departments, indicating that employee retention is not necessarily uniform throughout the organization.

Job Role Differences

Different job roles demonstrate different attrition patterns. This can help HR teams identify roles that may require further investigation.

Overtime and Attrition

The dashboard provides a direct comparison between overtime status and attrition, allowing HR management to investigate whether workload and overtime are associated with employee turnover.

Age and Attrition

The age-group analysis provides a demographic perspective on employee attrition and helps identify groups that may have different retention patterns.

Tenure and Attrition

Comparing years at the company with attrition helps HR understand how employee retention changes with organizational tenure.

Departmental Job Satisfaction

Differences in average job satisfaction between departments provide additional context for understanding employee experience.

8. Recommendations

Based on the dashboard analysis, the following recommendations can be considered:

1. **Monitor departments with higher attrition** and investigate the underlying reasons through employee feedback and HR analysis.
2. **Review high-attrition job roles** to identify potential issues related to workload, compensation, career progression, or job satisfaction.
3. **Monitor overtime patterns** and workload distribution to reduce the risk of employee burnout.

4. **Strengthen employee engagement initiatives** in departments where job satisfaction is comparatively lower.
 5. **Pay attention to employee tenure**, particularly during stages where attrition appears relatively higher.
 6. **Use the dashboard regularly for workforce monitoring** so HR teams can identify changing attrition patterns and take timely action.
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9. Business Value

The dashboard converts employee-level HR data into a **single interactive reporting view**.

Instead of manually examining individual employee records, HR stakeholders can quickly review:

- Workforce size
- Attrition
- Employee satisfaction
- Average age
- Compensation
- Employee tenure
- Department-level patterns
- Job-role patterns
- Overtime patterns
- Age-group patterns

The slicers further allow stakeholders to investigate specific workforce segments.

This makes the dashboard useful for **HR reporting, workforce monitoring, retention analysis, and management decision-making**.

10. Conclusion

The IBM HR Analytics project demonstrates the process of transforming raw HR data into an interactive Power BI dashboard.

The project involved **data preparation, KPI selection, dashboard planning, visualization, interactive filtering, analysis, insight generation, and recommendations**.

The final one-page dashboard provides a concise executive view of workforce characteristics and employee attrition patterns. Its interactive slicers allow users to explore the data from different departmental, demographic, job-role, job-level, and overtime perspectives.

Overall, the project demonstrates the ability to use Power BI to transform HR data into a **professional, business-oriented analytics solution** that can support workforce analysis and informed decision-making. This aligns with the ShadowFox Advanced Level expectations for interactive dashboards, executive metrics, visual storytelling, and decision-oriented recommendations.